

Device-Independent Quantum Communication Technology

From Theoretical Paradox to Hardware Security

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Agenda

- **The Security Gap:** Quantum Promise vs. Physical Reality
- **Theoretical Foundations:** EPR Paradox and Bell's Theorem
- **Hardware Architecture:** Engineering Entanglement
- **Vulnerabilities:** The APD Blinding Exploit
- **The Solution:** Device-Independent QKD (DI-QKD)
- **True Randomness:** QRNGs and Bias Extraction

The Quantum Promise vs. Physical Reality

- **Information-Theoretic Security:** Guaranteed by No-Cloning and Wave-function Collapse.
- **The Physical Gap:** Ideal proofs ignore fiber loss ($\alpha \approx 0.2$ dB/km) and hardware imperfections.
- **Security Collapse:** If detection hardware is compromised, the mathematical model of the quantum state becomes invalid.

Shor-Preskill Threshold

Secure key extraction is only possible if the Quantum Bit Error Rate (QBER) remains below $\approx 11\%$.

EPR Paradox and the CHSH Inequality

- **Local Realism:** Einstein's argument for "Hidden Variables" to avoid "spooky action".
- **CHSH Inequality (1964):** Tests correlations between Alice and Bob at varying angles.
- **Realism vs. QM:** Violation proves measurement actively defines reality.

Mathematical Bounds

Classical Hidden Variables: $S \leq 2$

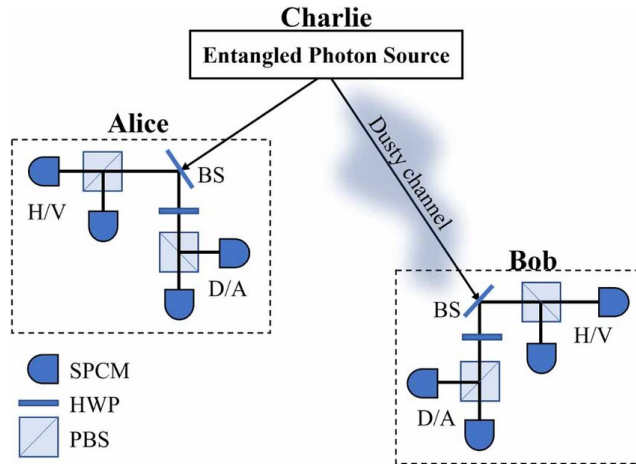
Tsirelson's Bound (Quantum): $S = 2\sqrt{2} \approx 2.828$

Bell State:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|HH\rangle + |VV\rangle)$$

Higher S values indicate stronger non-classical correlation.

Engineering Entanglement: The BBM92 Setup



The 4-Step Engineering Process

- 1 Generation:** SPDC via non-linear crystals (BBO/ppKTP).
- 2 Superposition:** Path overlap to erase "which-path" info.
- 3 Compensation:** Fixing phase errors via YVO_4 .
- 4 Splitting:** Wedge mirrors separate signal/idler.

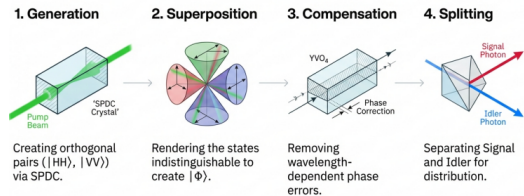
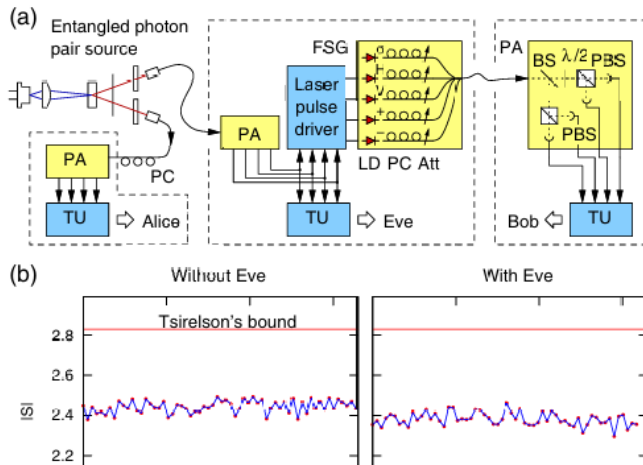


Figure: Entanglement Generation Process.

The Detector Blinding Exploit (Gerhardt et al.)



Device-Independent QKD: The Zero-Trust Model

- **Black Box Approach:** Treat all hardware as untrusted; security relies on statistical outputs.
- **Verification:** Use the Bell test directly on hardware. $S > 2$ ensures security.
- **Monogamy of Entanglement:** A third party (Eve) cannot correlate with a maximally entangled pair.

Paradigm Shift

Moves from modeling the **internal physics** of a device to verifying the **external correlations**.

Closing the Loophole: QRNGs

- **Unpredictability:** DI-QKD fails if Eve predicts measurement bases.
- **Quantum Randomness:** Spatial superposition in a 50/50 beam splitter.
- **Von Neumann Extractor:** Eliminates hardware bias by processing non-overlapping bit pairs.

Extractor Logic

Input: '01' → Output: 0

Input: '10' → Output: 1

Discard: '00' and '11'

Conclusion

- Physical layer vulnerabilities (APDs) compromise standard QKD.
- DI-QKD provides security even when devices are fundamentally compromised.
- Future networks must integrate robust Bell tests and bias-extracted QRNGs.

Thank You!

Questions?